

AI-Assisted Emergency Triage System Safety Considerations

Purpose

This document identifies key safety risks associated with the proposed AI-assisted emergency triage system at Mercer General Hospital and describes the mechanisms implemented to reduce those risks. The system includes both a nurse-facing Human–Computer Interaction (HCI) dashboard and a patient-facing Human–Robot Interaction (HRI) kiosk.

Human–Computer Interaction (HCI)

1. Alarm Fatigue from Repeated Non-Critical Alerts

Safety Consideration: Repeated notifications for low-risk patients during busy or overnight shifts may cause nurses to become desensitised, increasing the likelihood that a genuine emergency alert will be overlooked.

Safety Mechanism: The dashboard suppresses duplicate notifications and prioritises alerts based on clinical severity. Only ESI Level 1 and Level 2 patients generate high-priority visual alerts, while lower-priority cases use less intrusive indicators.

2. Reduced Display Legibility During High-Pressure Clinical Work

Safety Consideration: Small text, poor colour contrast, or cluttered screens may delay information recognition when nurses are making rapid triage decisions under stress.

Safety Mechanism: The interface uses a high-contrast colour scheme, a minimum 16-point font for critical information, simplified layouts, and colour-coded patient priority indicators so essential information can be identified immediately.

3. Over-Reliance on AI Recommendations

Safety Consideration: Nurses may accept AI recommendations without adequately reviewing the patient's condition, particularly during periods of high workload.

Safety Mechanism: Every recommendation includes a confidence score and an explanation of the clinical factors influencing the prediction. The system requires the nurse to actively accept or override the recommendation, and override decisions must include a documented justification recorded in the audit log.

Human–Robot Interaction (HRI)

1. Patient Injury from Unsafe Physical Interaction

Safety Consideration: Patients with reduced mobility or limited vision may accidentally collide with the kiosk or have difficulty positioning themselves to use integrated medical devices.

Safety Mechanism: The kiosk has rounded edges, a stable base, wheelchair-accessible height, non-slip flooring around the unit, and clearly marked standing positions to promote safe interaction.

2. Incorrect Voice Recognition in a Noisy Emergency Department

Safety Consideration: Background conversations, medical equipment, and ambulance activity may cause voice commands or symptom descriptions to be interpreted incorrectly, resulting in inaccurate patient information.

Safety Mechanism: The kiosk allows patients to switch between voice and touchscreen input, displays recognised responses for confirmation before submission, and provides an option to request immediate assistance from a triage nurse if communication problems occur.

3. Inaccurate AI Assessment Due to Sensor Failure

Safety Consideration: Faulty or disconnected devices such as the blood pressure cuff or pulse oximeter may provide incomplete or inaccurate vital signs, leading to an unreliable AI recommendation.

Safety Mechanism: The kiosk performs automatic device status checks before each assessment. If a sensor fails or produces invalid readings, the assessment is stopped, staff are notified, and the patient is directed to a triage nurse. The AI does not generate a recommendation until complete and valid data are available.

Summary

These safety mechanisms ensure that the AI-assisted triage system supports clinical decision-making without replacing healthcare professionals. The HCI dashboard reduces

cognitive workload while maintaining human oversight, and the HRI kiosk incorporates physical, communication, and technical safeguards to protect patients and ensure reliable data collection before AI-assisted triage.